

EXPERIMENT NO. 5

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TITLE

Simple Linear Regression – Study Hours vs Exam Score

OBJECTIVE

To implement simple linear regression on study-hours data by the regression equation and the standard (normal) equation, predict the score for 9 hours, and plot the regression line.

AIM

To fit $\hat{Y} = mX + b$ on hours-vs-score data with least squares and the normal equation, predict the 9-hour score, and evaluate with MSE and R^2 .

THEORY

Regression equation:

$$\hat{Y} = mX + b$$

Least-squares estimates:

$$m = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} \quad b = \bar{Y} - m\bar{X}$$

Prediction and goodness of fit:

$$\hat{Y}(9) = m \cdot 9 + b \quad \text{MSE} = \frac{1}{n} \sum (Y_i - \hat{Y}_i)^2 \quad R^2 = 1 - \frac{\sum (Y_i - \hat{Y}_i)^2}{\sum (Y_i - \bar{Y})^2}$$

PROCEDURE

1. Store hours (X) and scores (Y) of 8 students in NumPy arrays.
2. Compute m, b by least squares and by the normal equation.
3. Predict the score for 9 study hours with both equations.
4. Compute MSE and R^2 ; plot data points with the regression line.

DATASET DESCRIPTION

Dataset Name: Study hours vs exam score (given table)

Source: 8 students; X = hours (1–8), Y = score out of 100

Number of Samples: 8 Number of Features: 1 (hours)

Target / Class: Exam score (0–100)

Description: Y = [45, 50, 54, 60, 65, 70, 74, 80]; predict at X = 9.

PROGRAM

```
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt

X = np.array([1, 2, 3, 4, 5, 6, 7, 8], dtype=float)
Y = np.array([45, 50, 54, 60, 65, 70, 74, 80], dtype=float)

x_bar, y_bar = X.mean(), Y.mean()
m = ((X - x_bar) * (Y - y_bar)).sum() / ((X - x_bar) ** 2).sum()
b = y_bar - m * x_bar
print(f"Least squares: m = {m:.4f}  b = {b:.4f}")

Xb = np.c_[np.ones(len(X)), X]
t = np.linalg.inv(Xb.T @ Xb) @ Xb.T @ Y
print(f"Normal equation: b = {t[0]:.4f}  m = {t[1]:.4f}")

print(f"LS prediction for 9 hrs = {m * 9 + b:.2f}")
print(f"NE prediction for 9 hrs = {t[1] * 9 + t[0]:.2f}")

for nm, (mm, bb) in [("LS", (m, b)), ("NE", (t[1], t[0]))]:
    p = mm * X + bb
    mse = float(((Y - p) ** 2).mean())
    r2 = float(1 - ((Y - p) ** 2).sum() / ((Y - y_bar) ** 2).sum())
    print(f"{nm}: MSE={mse:.4f} R2={r2:.4f}")

plt.figure(figsize=(7, 4.5))
plt.scatter(X, Y, s=40, label="Data points")
xs = np.linspace(0, 10, 200)
plt.plot(xs, m * xs + b, linewidth=2, label="Regression line")
plt.plot(9, m * 9 + b, marker="*", markersize=14, label="Prediction (9 hrs)")
plt.xlabel("Hours study (X)")
plt.ylabel("Exam score (Y)")
plt.title("Simple Linear Regression: Study Hours vs Exam Score")
plt.legend()
plt.tight_layout()
plt.savefig("exp5_fit.png", dpi=150)
```

OUTPUT

Least squares: $m = 4.9762$ $b = 39.8571$
Normal equation: $b = 39.8571$ $m = 4.9762$
LS prediction for 9 hrs = 84.64
NE prediction for 9 hrs = 84.64
LS: $MSE=0.1845$ $R^2=0.9986$
NE: $MSE=0.1845$ $R^2=0.9986$

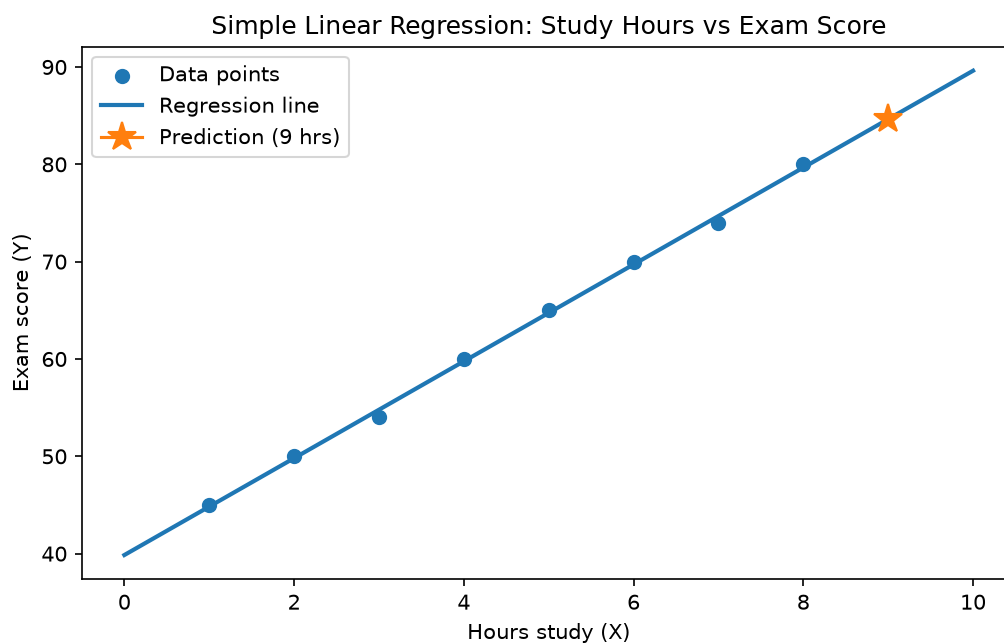


Fig. 1: Data points, regression line and 9-hour prediction.

RESULT

The above program was implemented and executed successfully using Python, and the corresponding output was obtained and verified. The results of the **least squares regression** were analyzed and found to be satisfactory, thereby achieving the desired objective of the experiment.

Faculty Signature: _____

Date: _____